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Agenda

- Bayes Theorem [conditional Probability]
- Dependent Events
- Independent Events

Assignments

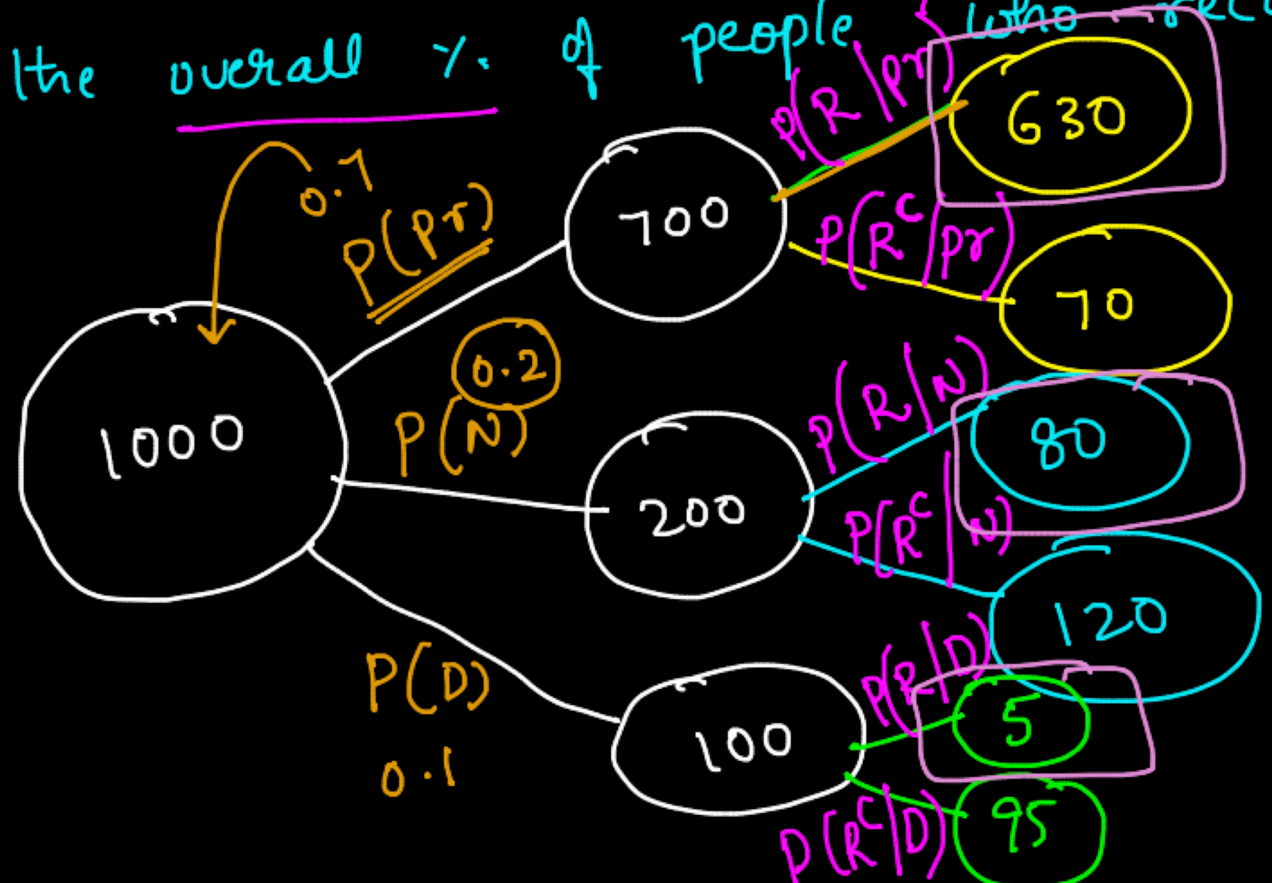
1. Conditional
2. law of total probability

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- ✓ 70% are promoters ✓
- ✓ 20% are neutral ✓
- ✓ 10% are detractors ✓

✓ 90% of promoters, 40% of neutral & 5% of detractors recommend product

(*) what is the overall % of people who recommend the product ?



$$\Rightarrow \frac{630 + 80 + 5}{1000}$$

$$= \frac{715}{1000} \Rightarrow 71.5\%$$

$$P(\text{Recom}) = P(Pr \cap \text{Recom}) + P(N \cap \text{Recom}) + P(D \cap \text{Recom})$$

$$= \frac{636}{1000} + \frac{80}{1000} + \frac{5}{1000} = \frac{715}{1000} = 71.5\%$$

$$P(Pr \cap N \cap D) = 0 \Rightarrow \text{Mutually Exclusive}$$

$$P(Pr \cup N \cup D) = 1 \Rightarrow \text{Exhaustive}$$

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(*) $P(A \mid B) = \dfrac{P(A \cap B)}{P(B)}$

(*) $P(C) = P(C \mid A) * P(A) + P(C \mid A^C) * P(A^C)$

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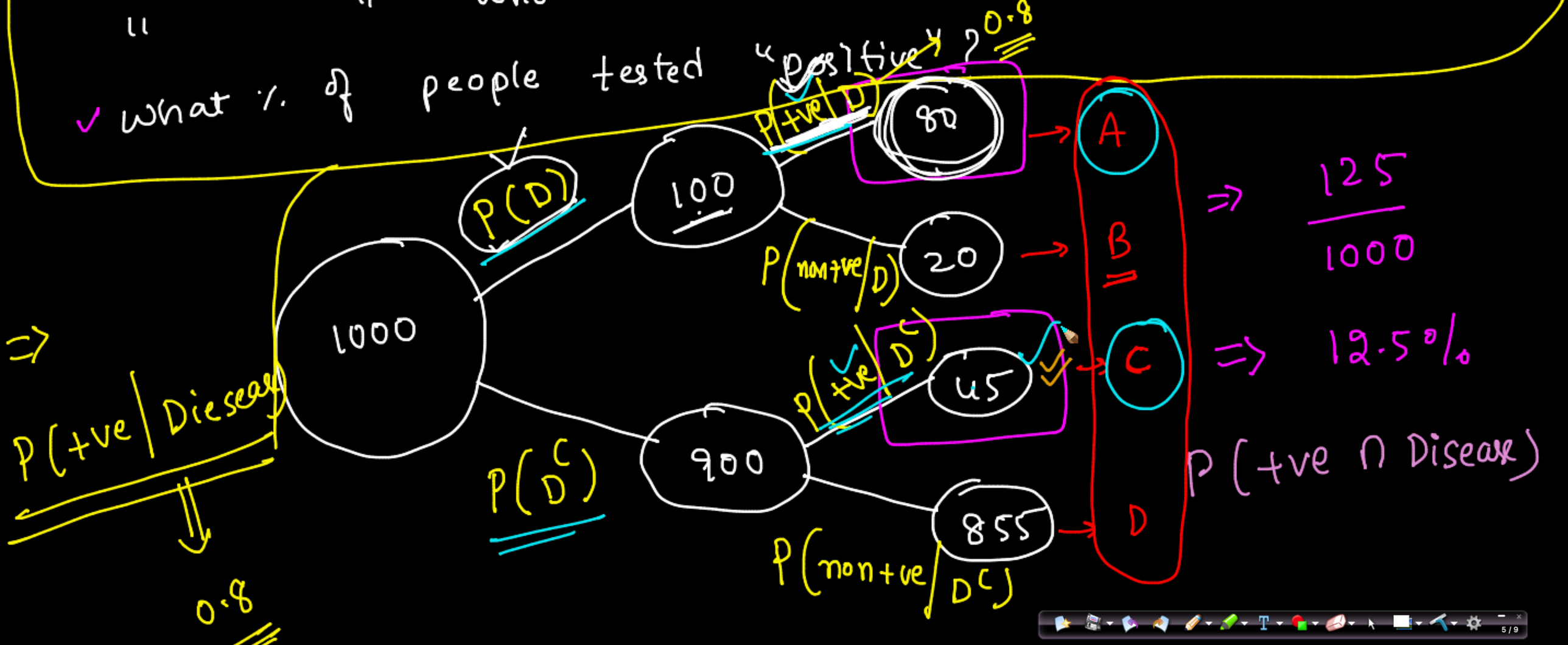
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A disease affects 10% of the population

Among those who have the disease, 80% get "positive"

" who don't have disease, 5% get "positive"

" what % of people tested "positive"?



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$$P(+ve \cap Disease) =$$

$$= P(D) * P(+ve \mid D)$$

$$= 0.1 * 0.8$$

$$= \underline{0.08}$$

$$\Rightarrow P(+ve \cap No\ Disease) = P(No\ Disease) * P(+ve \mid No\ Disease)$$

$$= 0.9 * 0.05$$

$$= \underline{0.045}$$

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⇒ Suppose you are tested +ve. What is the probability that you have the disease?

$P(D \mid +ve)$

a. $P(D)$

b. $P(+ve \mid D)$

✓ $P(D \mid +ve)$

d. $P(D \cap +ve)$

⇒ If you are a +ve, then you belong to (80 + 45) people. Among those how many have disease?



$= 0.64$

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$$P(D \mid +ve) = \frac{P(D \cap +ve)}{P(+ve)}$$

**

$$P(D \mid +ve) = \frac{P(D) * P(+ve \mid D)}{P(+ve)}$$

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$$P(D \mid +ve) = \frac{P(D \cap +ve)}{P(+ve)}$$

$$= \frac{P(D) * P(+ve \mid D)}{P(+ve)} = \frac{0.1 * 0.8}{0.125}$$

$$= 80/125 = 0.64$$

$$P(+ve) = P(D) * P(+ve \mid D) + P(D^c) * P(+ve \mid D^c)$$

$$= \frac{125}{1000}$$

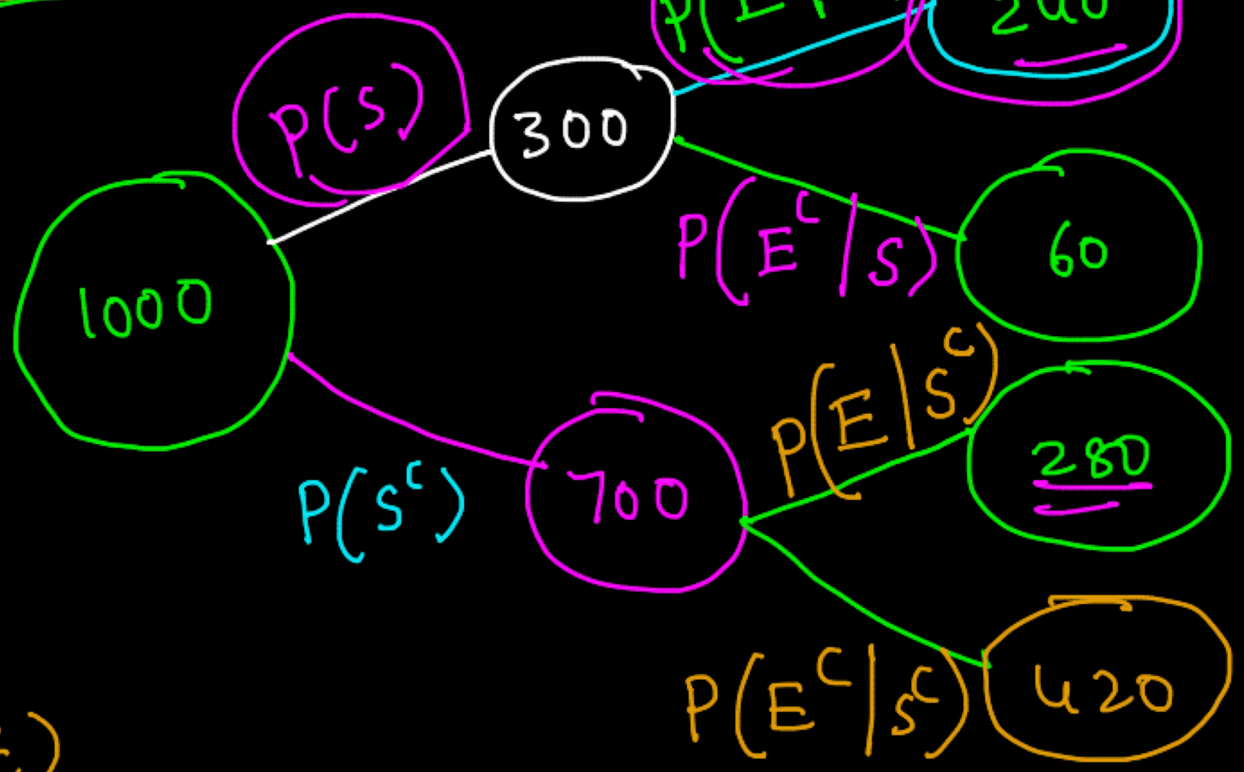
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30 % of the people know SQL ✓

80 % of the people who know SQL also know Excel

40 % of the people who don't know SQL also know Excel.

Among those who know Excel, what percentage know SQL?



$P(S|E)$

$\frac{240}{520}$

a) $P(S)$

b) $P(E|S)$

c) $P(S|E)$ ✓

d) $P(S \cap E)$

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$$P(S|E) = \frac{P(S \cap E)}{P(E)}$$

$$= \frac{P(S) * P(E|S)}{P(E)} \checkmark$$

$$P(E) = P(S) * P(E|S) + P(S^c) * P(E|S^c)$$

$$= 0.3 * 0.8 + 0.7 * 0.4$$

$$= 0.24 + 0.28 \Rightarrow 0.52$$

$$= \frac{0.24}{0.52}$$

$$= \frac{24}{52}$$

$$= 0.46 \checkmark$$

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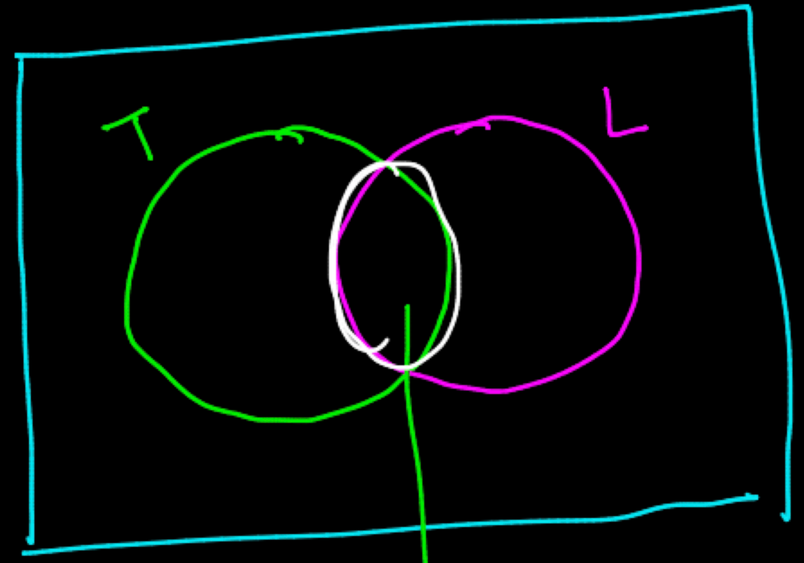
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7% people are on Twitter

5% people are on LinkedIn

4% people are on both LinkedIn & Twitter

Random person is choosen. what is the probability that he is on Twitter?



random linked in user is choosen:

$P(T) = 7/100$

$= 0.07$

$\Rightarrow P(T \mid L) =$

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$$P(T|L) = \frac{P(T \cap L)}{P(L)} = \frac{0.04}{0.05}$$

$$= \frac{4}{5} = 0.8$$

** The extra inf that a person is on LinkedIn, did this ↑ (or) ↓ his probability of being on twitter?

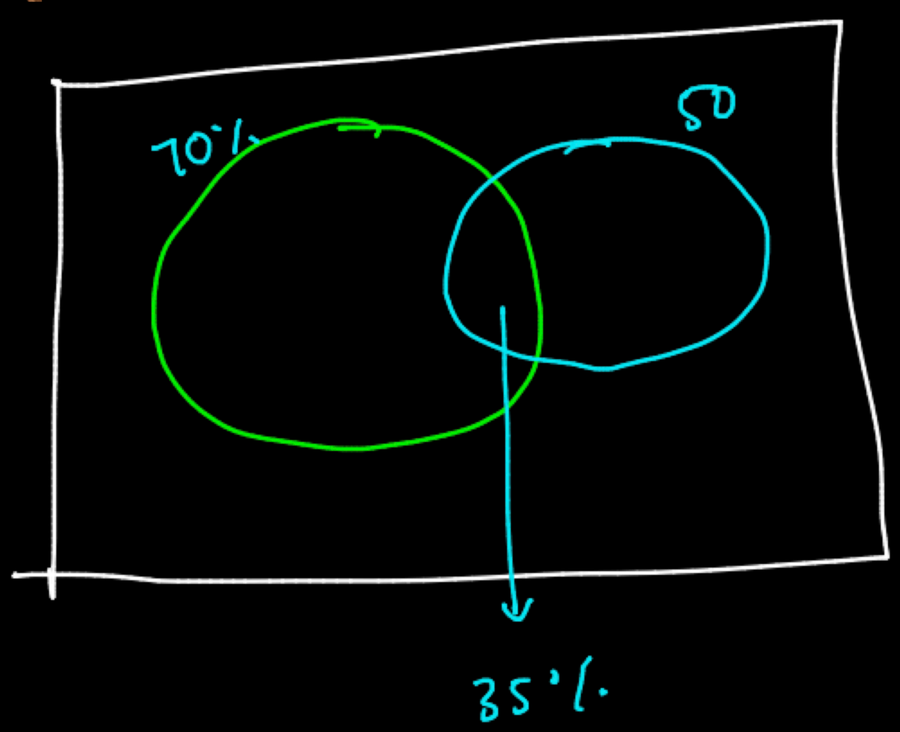
Dependent Event

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Among those who saw the ad 70% saw it on Youtube, 50% saw it on Amazon

35% saw it on both

Random person : probability that he saw the ad in Youtube



$P(Y) = 0.7$ ✓✓

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random person who saw the ad on Amazon is choosen.
probability that he also saw the ad on Youtube.

$$P(Y \mid A) = \frac{P(A \cap Y)}{P(A)}$$

Independent Events

$$= \frac{0.35}{0.5} = \frac{\cancel{35}\ 7}{\cancel{50}\ 10}$$

$P(Y)$

$$= 0.7$$

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If A and B are independent

$$P(A|B) = P(A)$$

$$P(B|A) = P(B)$$

$$\frac{P(A \cap B)}{P(B)} = P(A)$$

$$P(A \cap B) = P(A) * P(B)$$

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Toss & Dice

$$S = \{H1, H2, H3, H4, H5, H6, T1, T2, T3, T4, T5, T6\}$$

A = Coin is head

$$A = \{H1, H2, H3, H4, H5, H6\}$$

$$P(A) = \frac{6}{12} = \frac{1}{2}$$

$$P(A \cap B) = \frac{1}{12}$$

$\{H3\}$

B = Dice is 3

$$B = \{H3, T3\}$$

$$P(B) = \frac{2}{12} = \frac{1}{6}$$

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$P(A \mid B) = P(A) \Rightarrow$ A & B

$\Downarrow$

independent

$1/2$

$$\frac{P(A \cap B)}{P(B)} \Rightarrow \frac{1/4}{1/2}$$

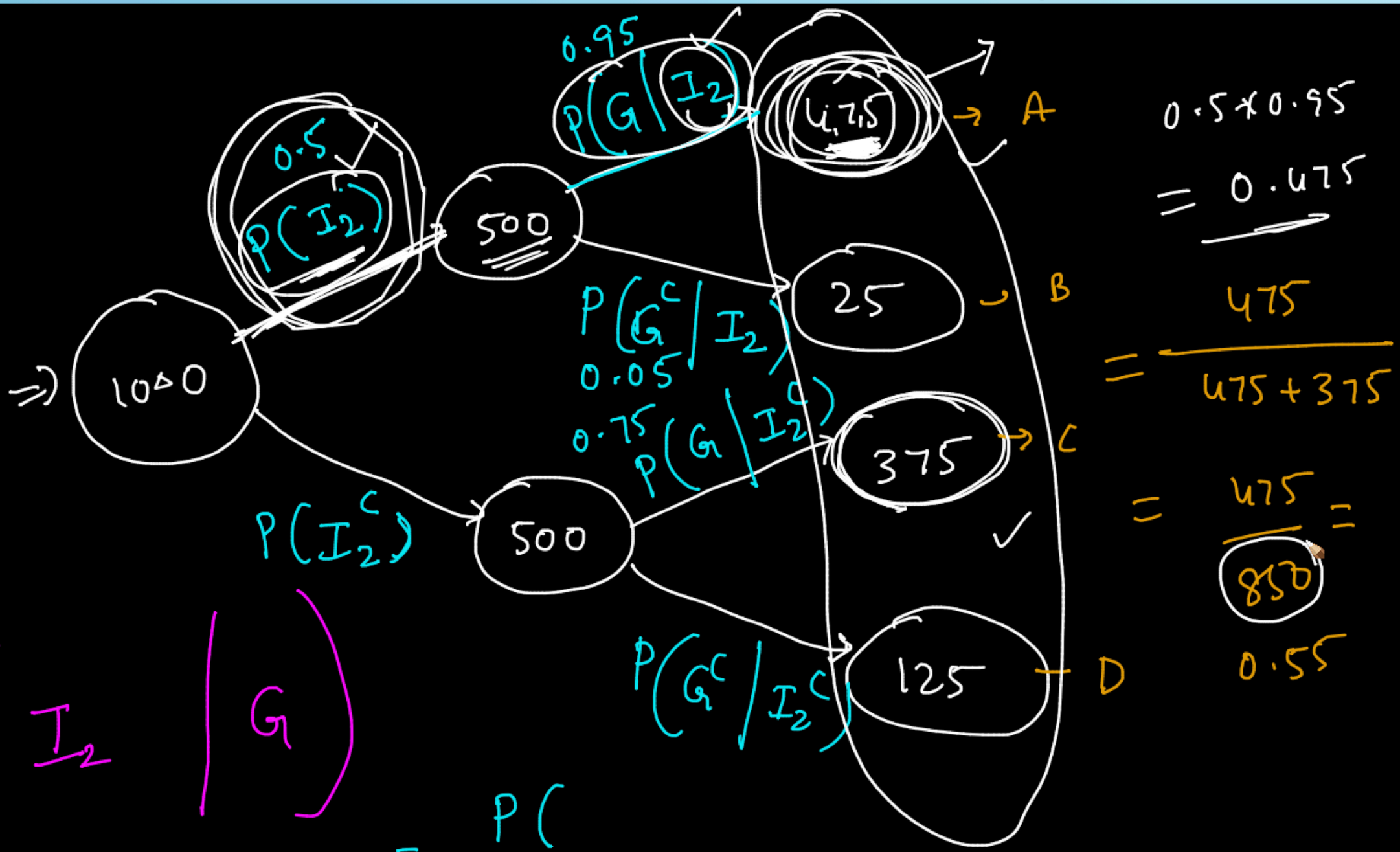
$$\Rightarrow 1/2$$

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Interview problem (Amazon)

⇒ 50% of the people who gave the first round we called for the second round. 95% of the people who got invited for the second round felt that they had a good first round. 75% of the people who did not get invited for the second round also felt that they had a good first round. Given that a person felt good about the first round, what is the probability that he cleared the first round?

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$$P\left(I_2 \mid G\right) = P($$

$$0.5 \times 0.95 = 0.475$$

$$= \frac{475}{475 + 375}$$

$$= \frac{475}{850} = 0.55$$

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$$P(I_2 \mid G) = \frac{P(I_2 \cap G)}{P(G)}$$

$$= \frac{P(I_2) * P(G \mid I_2)}{P(G)}$$

$$= \frac{0.5 * 0.95}{0.5 \times 0.95 + 0.5 * 0.75} =$$

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$$P(A|B) = \frac{P(A \cap B)}{P(B}$$